

Ve 216: Introduction to Signals and Systems

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Shanghai Jiao Tong University

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Based on Lecture Notes by Prof. Jeffrey A. Fessler and Prof. Yong Long

Outline

- 1 1. Signals & Systems (Fundamentals)
 - Overview
 - Signal and System Definition
 - Classification of Signals
 - Signal Notation
 - Transformations of CT signals
 - Signal Characteristics
 - Exponential signals
 - Singularity functions (1.4)
 - Continuous-time systems
 - Summary

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Overview

Signals

- definition
- classes
- notation
- transformations
(operations)
- important signals
(Skip: 1.3.2, 1.3.3, 1.4.1)

Systems

- definition
- block diagrams
- system interconnection
- classes
- **linearity**, **time-invariance**

Goal: eventually system design; must first learn to analyze!

Overview

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- classes
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Signal Definition

Definition

A **signal** is any “physical” quantity that varies with time or space (or any other independent variable or variables).

Often when we discuss signals we refer to **mathematical representation** of the physical quantity.

Example

An approaching **ambulance siren** produces a time-varying change in acoustic pressure that our ears perceive as sound.

$$s(t) = (1 + t) \sin \left(2\pi [1000t + 10t^2 + 300 \sin(2\pi t/2)] \right)$$

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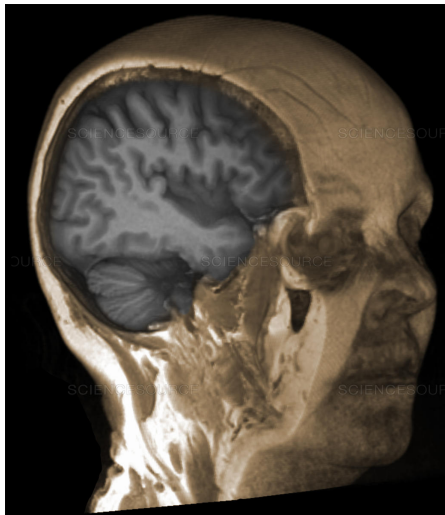
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Example: Ambulance Siren

$$s(t) = (1 + t) \sin\left(2\pi[1000t + 10t^2 + 300 \sin(2\pi t/2)]\right)$$

- The $(1 + t)$ **amplitude term** represents increasing loudness as the ambulance approaches.
- The $1000t$ term represents the 1kHz **siren oscillation**.
- The $10t^2$ term represents **increasing pitch** due to the **Doppler effect** as the ambulance approaches.
- The $300 \sin(2\pi t/2)$ term represents **the eeh-ooh-eeh-ooh periodic variation in pitch**.

Example: 3D Magnetic Resonance Imaging (MRI)



- The signal can be a function of **time** or **space** or both.
- The signal can be of **high-dimension**.

System Definition

Definition

A **system** is a physical “device” that performs an operation on a signal.

Example

The human ear converts acoustic signals into electrical nerve synapses (another signal) that are processed by the brain. The input and output signals are different physical quantities.

System Definition

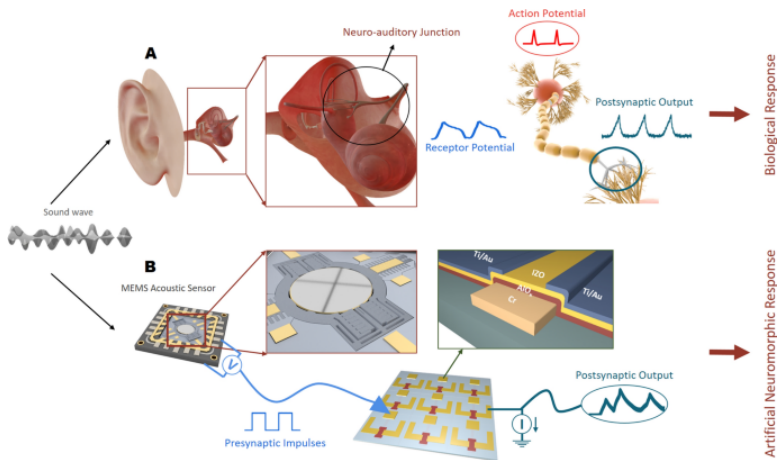
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Ear: An amazing acoustic system



Signal processing

One of the main roles of electrical engineers is to **design and analyze systems** that take some **input signal** and produce some related (but almost always different) **output signal**.

We refer such operations as **signal processing**.

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Signals and Systems

Example

For an **audio amplifier**, ideally the output signal is “simply” an amplified version of the input signal. (On paper it is easy:

$$s_{\text{out}}(t) = as_{\text{in}}(t).$$

But implementing this in analog hardware with **minimal distortion** is **nontrivial**.)

This course will emphasize **continuous-time** or **analog** signals, and briefly introduce **discrete-time** or **digital** signals at the end.

(Portions of Chapters 1-10)

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A signal is a function

Mathematically, a **signal** is a **function** of one or more independent variables.

Question

What is a function?

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It is a rule that assigns to each value in the range set at least one of the values in the domain set.

Classification of Signals: Dimensionality (1)

One way to classify signals is by the dimension of the **domain of the function**, *i.e.* how many arguments the function has.

Definition

A **one-dimensional** signal is a function of a single variable, *e.g.* time.

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What is a ***M-dimensional*** signal?

A ***M-dimensional*** signal is a function of M independent variables.

Classification of Signals: Dimensionality (3)

Example

A sequence of BW TV pictures $I(x, y, t)$ is a scalar valued function of two spatial coordinates x and y and time t , so it is a **3D signal**.

We will focus on **one-dimensional** signals in this course, generally considering the independent variable to be **time t** .

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Another way to classify signals is by the **dimension of the range** of the function, *i.e.*, the space of values the function can take.

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A **scalar** or **single-channel** signal is a function of a real-valued scalar or complex-valued scalar.

Question

What is a **multichannel** signal?

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Classification of Signals: Dimensionality (5)

Example

A **color** TV picture can be described by a red, blue and green signal, whereas a **BW** TV picture is scalar valued.

We will focus on **scalar** signals in this course, both real and complex.

Most of the design/analysis techniques **generalize to multichannel and multidimensional** signals.

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Continuous-time signals

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A **continuous-time signal** or **analog signal** is a function defined for all times $t \in (-\infty, \infty)$, or at least over some continuous interval (a, b) .

Example

$$x(t) = e^{-t^2}, \quad -\infty < t < \infty. \text{ (Picture)}$$

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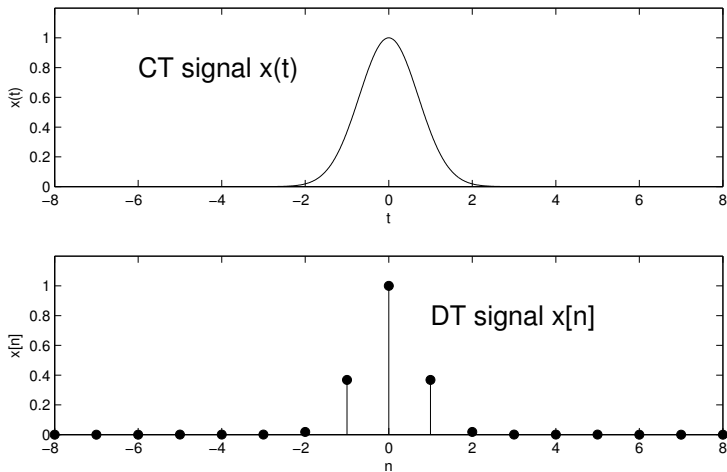
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Continuous-time signals vs. discrete-time signals



Classification of Signals: time characteristics

Classify signals by **time characteristics**

- 1 **Continuous-time** signals or **analog** signals
- 2 **Discrete-time** signals

Discrete-time signals arise from

- **Sampling** a continuous signal at discrete time instants
- **Accumulating** a quantity over a period of time

Example

When counting number of heart attacks per month, n would index the month, and $x[n]$ would be the number.

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Classification: Value characteristics

Definition

A **continuous-valued signal** or **continuous-amplitude signal** can take any value in some continuous interval.

Example

Voltage between 0 and 5 volts.

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- 2 **Random signals** evolve over time in an **unpredictable** manner.

Example

“Hiss” or “noise” in an audio system.

We will focus on **deterministic signals**, although reducing noise (eliminating a random component) is often a goal in designing signal processing systems.

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Classification: Our focus

We will focus on

- single-channel, one-dimensional, continuous-valued, continuous-time signals.
- $x(t)$ is a scalar valued function of a real independent variable.
- Mathematically

$$x : \mathbb{R} \rightarrow \mathbb{R} \text{ or } x : \mathbb{R} \rightarrow \mathbb{C}$$

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Signal notation (mathematical representation)(1)

1 Graphically (**Picture**):

2 Braces or piecewise notation: $x(t) = \begin{cases} e^{-t}, & t > 0 \\ e^t, & t \leq 0. \end{cases}$

3 Formula: $x(t) = e^{-|t|}$.

4 In terms of other functions: $x(t) = s(t) + s(-t)$ where

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5 Fourier representation:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2}{1 + \omega^2} e^{j\omega t} d\omega$$

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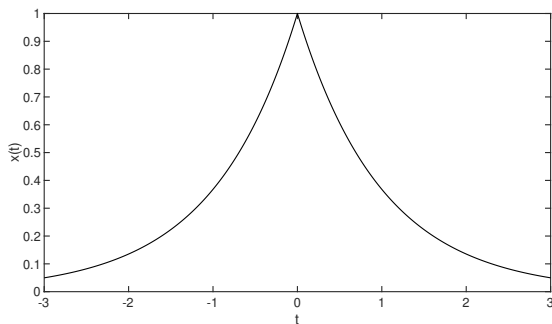
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Eventual goal

- **Eventual goal:** analyze **interesting signals** and to analyze and design **useful systems**.
- Such signals and systems consist of combinations of **simpler (less interesting?)** signals and systems.
- So we walk before we run...

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Transformations of CT signals

- Time transformations
 - Folding/reflection/time-reversal
 - Time-scaling
 - Time-shifting/time-delay
 - General time transformation
- Amplitude transformations
 - Amplitude reversal
 - Amplitude scaling
 - Amplitude shifting
- More signal operations
 - Differentiator
 - Integrator
- Operations with two signals

Change of variables

If $x(t) = e^{-(t-2)}$ then $y(t) = x\left(\frac{t-1}{3}\right)$ is another function;

$$y(t) = e^{-[(t-1)/3-2]} = e^{-\left(\frac{t-7}{3}\right)}.$$

In calculus, this type of transformation is called a **change of variables**.

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Time transformations

Here we give some new names to such transformations to reflect the **physical meaning** of the mathematics.

Example

$$x(t) = \begin{cases} e^{-(t-2)}, & t \geq 2, \\ 0, & \text{otherwise.} \end{cases} \quad \textit{(Picture)}.$$

(used throughout)

One can apply time transformations both **graphically** and **mathematically**. Both approaches are useful.

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Folding/reflection/time-reversal: $y(t) = x(-t)$ (1)

Folding/reflection/time-reversal

$$y(t) = x(-t)$$

Example

- Backwards play a movie/audio tape.
- A mirror is an optical system that does “space reversal”.

Example

Find $y(t) = x(-t)$ for $x(t)$ above.

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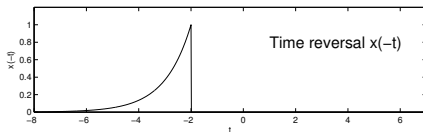
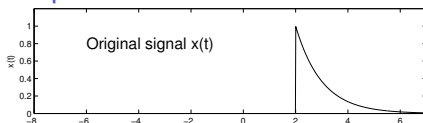
Folding/reflection/time-reversal: $y(t) = x(-t)$ (2)

Mathematical method:

- Replace all t 's with $-t$,
- Simplify where possible.

$$\begin{aligned}
 y(t) &= x(-t) \\
 &= \begin{cases} e^{-(-t-2)}, & -t \geq 2, \\ 0, & \text{otherwise} \end{cases} \\
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Graphical method:



Note that $y(t)$ becomes a “mirror image” of $x(t)$ around $t = 0$.

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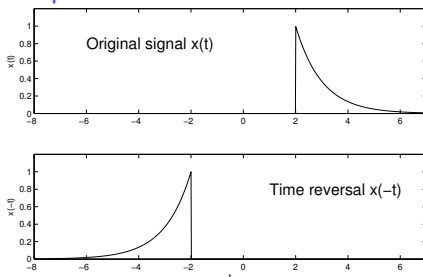
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Time-scaling: $y(t) = x(at), a > 1$

Time-scaling

$$y(t) = x(at)$$

$a > 1$ will shrink or compress the signal

Example

Playing a recording at 3 times the normal speed.

Example

Find $y(t) = x(3t)$ for $x(t)$ above.

Time-scaling: $y(t) = x(at)$, $a > 1$

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$$y(t) = x(3t) = \begin{cases} e^{-(3t-2)}, & 3t \geq 2 \\ 0, & \text{otherwise} \end{cases} = \begin{cases} e^{-3(t-2/3)}, & t \geq 2/3 \\ 0, & \text{otherwise} \end{cases}$$

Time-scaling: $y(t) = x(at)$, $a < 1$

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slow-motion part of a movie

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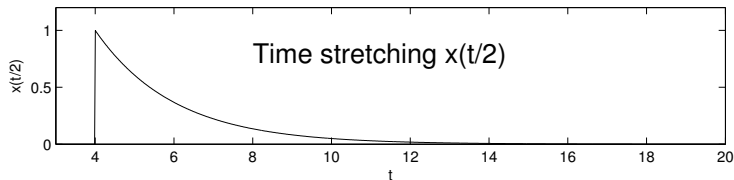
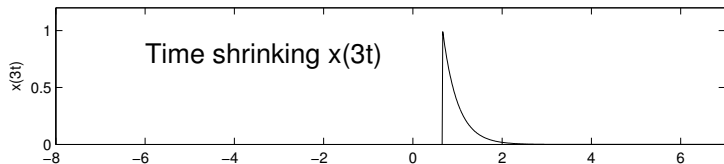
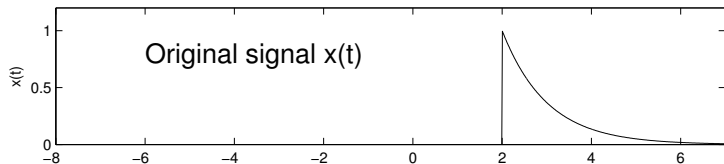
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Find $y(t) = x(t/2)$ for $x(t)$ above.

$$y(t) = x(t/2) = \begin{cases} e^{-(t/2-2)}, & t/2 \geq 2 \\ 0, & \text{otherwise} \end{cases} = \begin{cases} e^{-(t-4)/2}, & t \geq 4 \\ 0, & \text{otherwise} \end{cases}$$

Time-scaling: $y(t) = x(at)$



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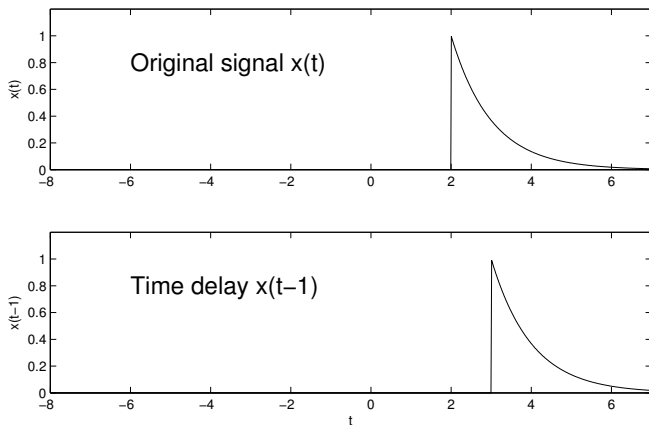
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Find $y(t) = x(t - 1)$ for $x(t)$ above.

$$y(t) = x(t-1) = \begin{cases} e^{-(t-1-2)}, & t-1 \geq 2 \\ 0, & \text{otherwise} \end{cases} = \begin{cases} e^{-(t-3)}, & t \geq 3 \\ 0, & \text{otherwise} \end{cases}$$

Time shifting: $y(t) = x(t - t_0)$ (Cont.)



Note that $x(t-1)$ *delays* the signal, which means it *shifts to the right* (starts later in time).

General time transformation

General time transformation involves all three of the above time transformations (time reversal, time scaling, and time shifting).

Two distinct (but related) forms:

$$y(t) = x(at - b) = x\left(\frac{t - t_0}{w}\right)$$

where $t_0 = b/a$ and $w = 1/a$ or
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Mathematical time transformation

mathematical recipe:

- 1 Replace **all** occurrences of t in the definition of $x(t)$ with $at - b$ or with $\frac{t-t_0}{w}$.
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Find $y(t) = x(-t/2 + 5) = x\left(\frac{t-10}{-2}\right)$ for $x(t)$ above.

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$$x(t) = \begin{cases} e^{-(t-2)}, & t \geq 2, \\ 0, & \text{otherwise.} \end{cases}$$

$$\begin{aligned} y(t) &= x(-t/2 + 5) \\ &= \begin{cases} e^{-(-t/2+5-2)}, & -t/2 + 5 \geq 2 \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} e^{(t-6)/2}, & t \leq 6 \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

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Graphical time transformations

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*How to perform a general time transformation **graphically**?
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The answer is: it depends on which of the two above forms you use to write the transformation!

Form 1: $y(t) = x(at - b)$

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- Introduce an intermediate signal $d(t) = x(t - b)$.
- Clearly $d(t)$ is just a **delayed** version of $x(t)$ by amount b .
- But $y(t) = d(at)$, which is just a **scaled** version of the signal $d(t)$.

To find $y(t) = x(at - b)$ graphically we must

- 1 **time-delay** the signal $x(t)$ by b .
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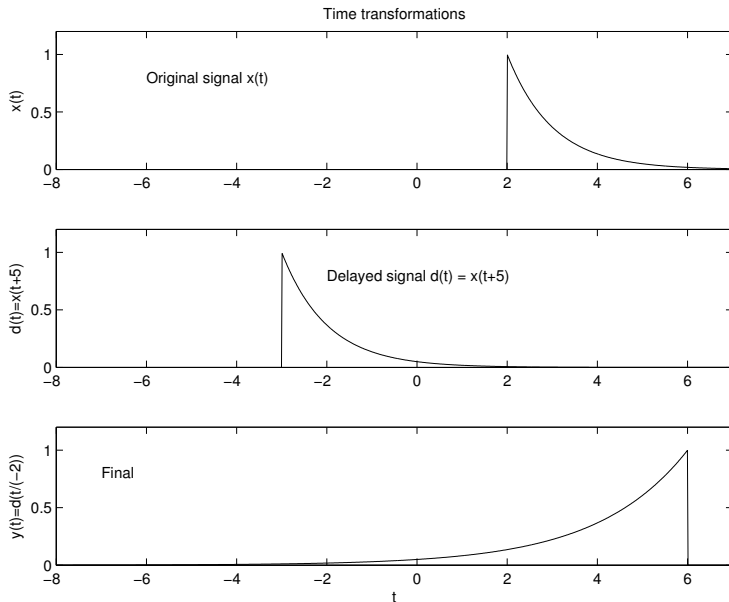
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Form 1: $y(t) = x(at - b)$ (Cont.)



Form 2: $y(t) = x\left(\frac{t-t_0}{w}\right)$

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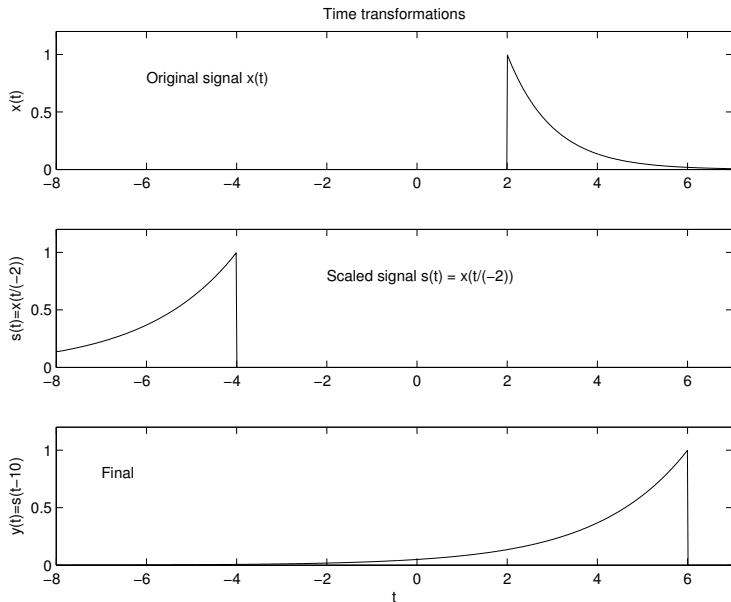
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Amplitude transformations

- ① **amplitude reversal** $y(t) = -x(t)$
- ② **amplitude scaling** $y(t) = ax(t)$
- ③ **amplitude shifting** $y(t) = x(t) + b$

Example

(Using all three.) Find $y(t) = -3x(t) + 2$ for the $x(t)$ above.

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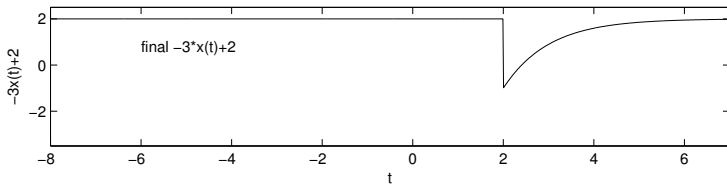
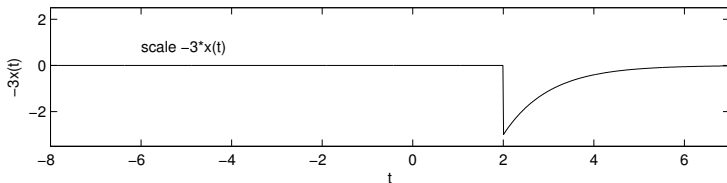
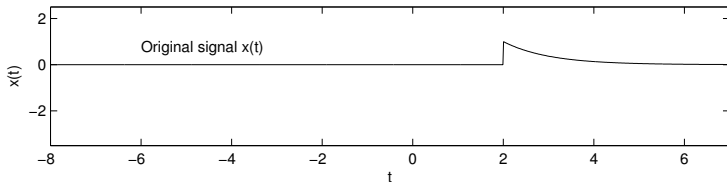
Example

(Using all three.) Find $y(t) = -3x(t) + 2$ for the $x(t)$ above.

$$y(t) = -3x(t) + 2 = \begin{cases} -3e^{-(t-2)} + 2, & t \geq 2 \\ -3 \cdot 0 + 2, & \text{o.w.} \end{cases}$$

Amplitude transformations

Graphically: scale amplitude by -3 and then shift up by 2.



Outline

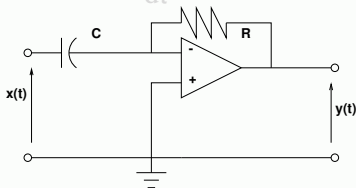
- 1 1. Signals & Systems (Fundamentals)
 - Overview
 - Signal and System Definition
 - Classification of Signals
 - Signal Notation
 - **Transformations of CT signals**
 - Signal Characteristics
 - Exponential signals
 - Singularity functions (1.4)
 - Continuous-time systems
 - Summary

Differentiator $y(t) = \frac{d}{dt}x(t)$

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$$y(t) = -RC \frac{d}{dt}x(t)$$



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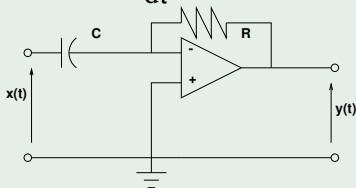
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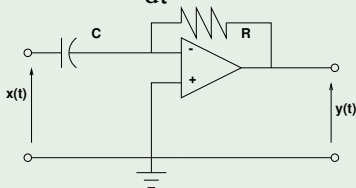
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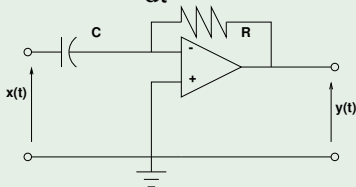
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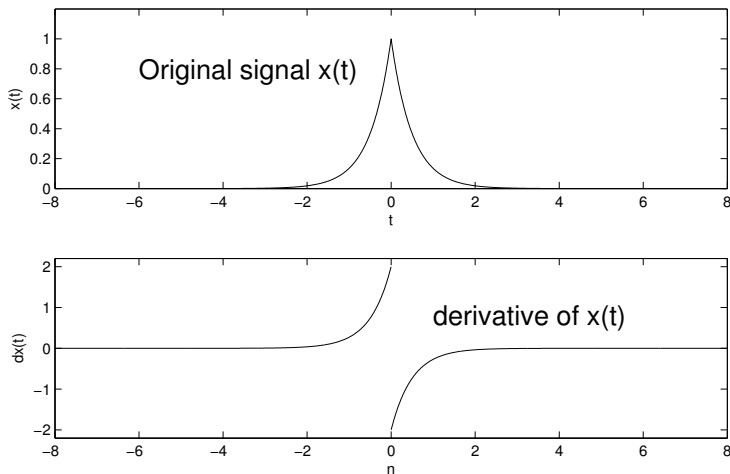
Example

Find the differentiated signal of $x(t) = e^{-2|t|}$.

$$\begin{aligned} y(t) &= \frac{d}{dt}x(t) \\ &= \begin{cases} -2e^{-2t}, & t > 0 \\ 2e^{2t}, & t < 0 \\ \text{?}, & t = 0. \end{cases} \end{aligned}$$

The derivative of $x(t)$ is *undefined* at $t = 0$.

Differentiator of $x(t) = e^{-2|t|}$

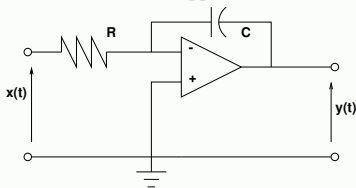


Integrator: $y(t) = \int_{-\infty}^t x(\tau) d\tau$

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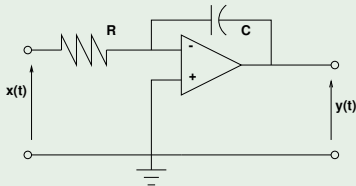
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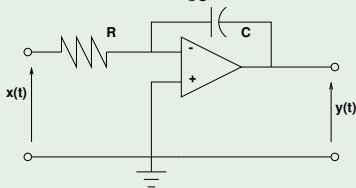
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Find the integrated signal of $x(t) = e^{-|t|}$.

Integrator: example

Solution

- 1 rewrite $x(\cdot)$ in terms of τ

$$x(\tau) = \begin{cases} e^{-\tau}, & \tau > 0 \\ e^{\tau}, & \tau \leq 0. \end{cases}$$

- 2 For $t \leq 0$:

$$y(t) = \int_{-\infty}^t e^{\tau} d\tau = e^t$$

- 3 For $t \geq 0$:

$$\begin{aligned} y(t) &= \int_{-\infty}^0 e^{\tau} d\tau + \int_0^t e^{-\tau} d\tau \\ &= e^0 + (-e^{-\tau}) \Big|_0^t = 1 + (1 - e^{-t}) \\ &= 2 - e^{-t}. \end{aligned}$$

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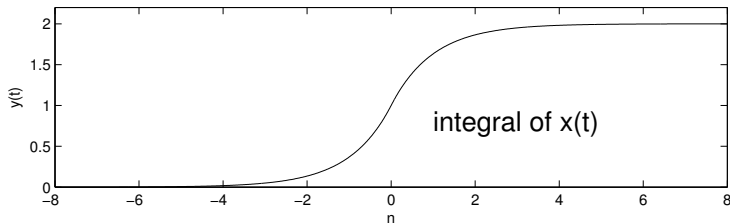
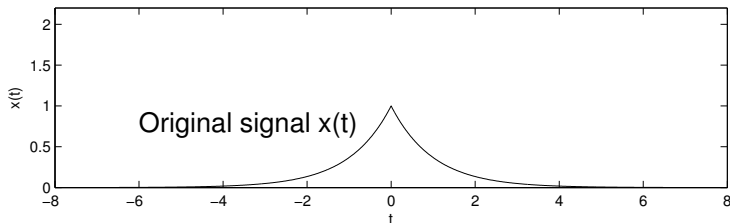
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Integrator: example (Cont.)

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Integrator vs. Integration in calculus

Question

What is the *distinction* between *simple integration* of the kind learned in *calculus* (computing area under a curve) and the *integrator system* described here.

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What is the *distinction* between *simple integration* of the kind learned in *calculus* (computing area under a curve) and the *integrator system* described here.

- When you calculate “area under a curve” you compute a *single number* from the curve, e.g. $\int_0^1 t^2 dt = 1/3$.
- *Integrator system* in signals and systems transforms one function $x(t)$ into another *function* $y(t)$.
It is a “running integral” whose upper limit is t .

Outline

1. Signals & Systems (Fundamentals)

- Overview
- Signal and System Definition
- Classification of Signals
- Signal Notation
- Transformations of CT signals
 - Time transformations
 - Amplitude transformations
 - More signal operations
 - Operations with two signals
- Signal Characteristics
 - Periodic/aperiodic signals
 - Even and odd signals
 - Energy and power signals
- Exponential signals
- Singularity functions (1.4)
 - Unit step signal
 - Rect(angle) function
 - Unit impulse function $\delta(t)$ (1.4.2, 2.5)
- Continuous-time systems

Operations with two signals

Operations with two signals

- ① **sum** of two signals $y(t) = x_1(t) + x_2(t)$
- ② **product** of two signals $y(t) = x_1(t) x_2(t)$.

Add or multiply two signals **at every time point**.

Example

If $x_1(t) = e^{-(t-3)^2}$, $x_2(t) = \sin(13t)$, then
 $y(t) = x_1(t)x_2(t) = e^{-(t-3)^2} \sin(13t)$.

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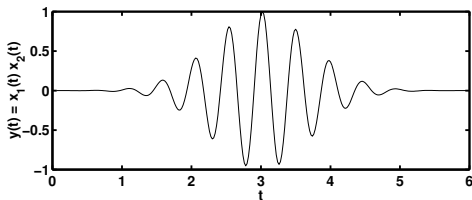
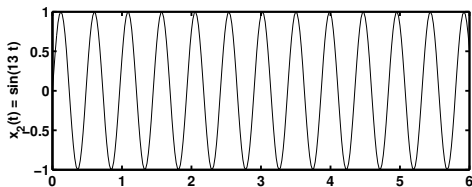
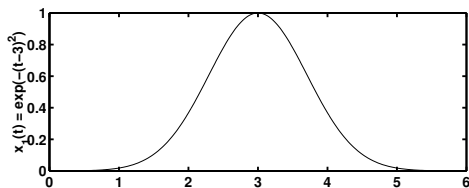
Example

If $x_1(t) = e^{-(t-3)^2}$, $x_2(t) = \sin(13t)$, then

$y(t) = x_1(t)x_2(t) = e^{-(t-3)^2} \sin(13t)$.

This is called **amplitude modulation**.

amplitude modulation



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Periodic signals

Why study **periodic signals**?

- important for analysis
- solution to ideal LC electrical circuits
- periodic physical phenomena: frictionless pendulums, earth rotation, heart rhythms, etc.

Definition

$x(t)$ is **periodic** with a **period** $T > 0$ iff

$$x(t + T) = x(t) \quad \forall t \quad (1)$$

Definition

If no such $T > 0$ exists, $x(t)$ is called **aperiodic**

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Fundamental period

Definition

The **fundamental period** T_0 of a signal is the smallest value of T satisfying (1).

Theorem

A signal that is periodic with period $T > 0$, is also periodic with period nT for any integer $n \neq 0$, i.e.

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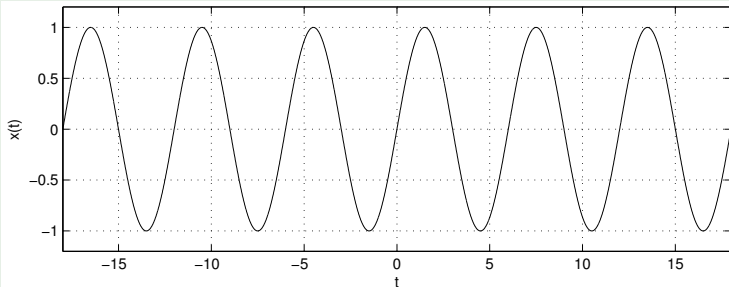
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Periodic signals: example

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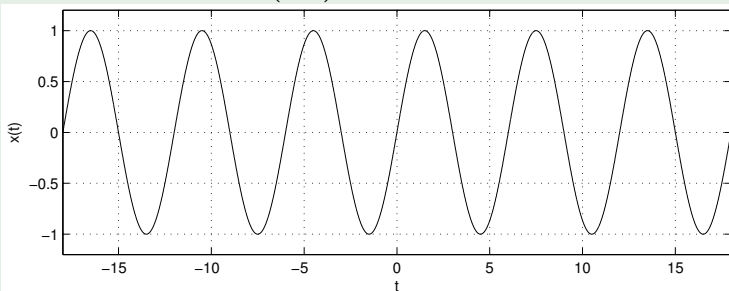
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Periodic signals: example

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$x(t) = \sin\left(\frac{\pi}{3}t\right) = \sin\left(\frac{2\pi}{T_0}t\right)$ What is fundamental period?



$$T_0 = 6 = \frac{2\pi}{\pi/3}.$$

Sums of two periodic signals

Question

Suppose $x_1(t)$ is periodic with period T_1 and $x_2(t)$ is periodic with period T_2 and $x(t) = x_1(t) + x_2(t)$.

- Is $x(t)$ periodic?*
- If so, what is a period T of $x(t)$?*

Sums of two periodic signals: solution (1)

Solution

- 1 *Easy case:* if $T_1 = T_2$ then, $x(t)$ is periodic, and $T = T_1 = T_2$.
- 2 *General case.* We know $x_1(t) = x_1(t + T_1)$ and $x_2(t) = x_2(t + T_2)$ and $x(t) = x_1(t) + x_2(t)$.
We want to determine if there is any value of $T > 0$ such that $x(t) = x(t + T)$.

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Sums of two periodic signals: solution (2)

Suppose there is a value of $T > 0$ that satisfies $T = n_1 T_1$ and $T = n_2 T_2$, for some nonzero integers n_1 and n_2 . Then

$$\begin{aligned}x(t + T) &= x_1(t + T) + x_2(t + T) = x_1(t + n_1 T_1) + x_2(t + n_2 T_2) \\&= x_1(t) + x_2(t) = x(t),\end{aligned}$$

so $x(t)$ is periodic with period T .

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Sums of two periodic signals: solution (3)

The conditions $T = n_1 T_1$ and $T = n_2 T_2$ are equivalent to requiring that

$$n_1 T_1 = n_2 T_2 \text{ so } T_1/T_2 = n_2/n_1,$$

which means that T_1/T_2 is **rational**, a ratio of two integers.

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A sum of two periodic signals is periodic iff the ratio of their periods is rational.

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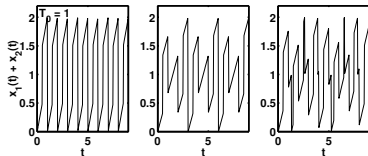
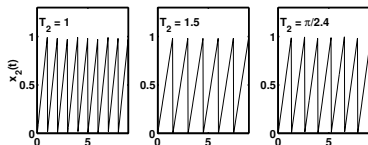
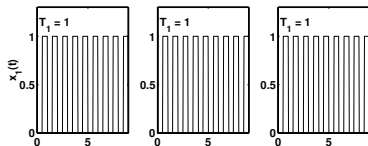
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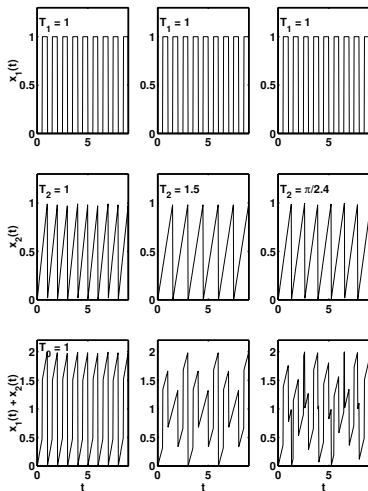
Sums of two periodic signals: example

Are the signals in the third row periodic?



Sums of two periodic signals: example

Are the signals in the third row periodic?



Yes. Yes. No.

Sums of two periodic signals: fundamental period

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The least common multiple of the fundamental periods of the two signals is a period of the sum. Is it the fundamental period?

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The least common multiple of the fundamental periods of the two signals is a period of the sum. Is it the fundamental period?
In general it may not be the fundamental period.

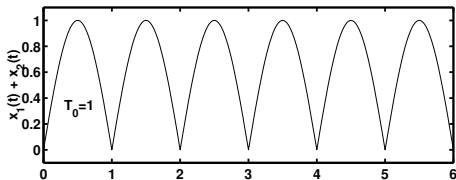
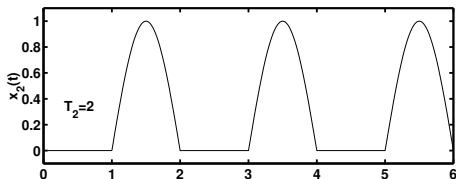
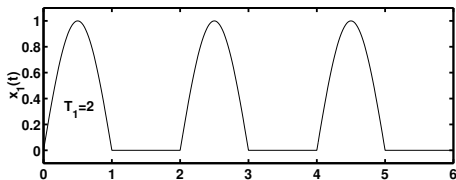
Sums of two periodic signals: fundamental period

Question

*The least common multiple of the fundamental periods of the two signals is a period of the sum. Is it the fundamental period?
In general it may not be the fundamental period.*

To find a least common multiple, you find the smallest values of n_1 and n_2 such that $n_1 T_1 = n_2 T_2$.

Sums of two periodic signals: fundamental period



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Even and odd symmetry

Definition

$x(t)$ has **even symmetry** iff $x(-t) = x(t) \forall t$

Definition

$x(t)$ has **odd symmetry** iff $x(-t) = -x(t) \forall t$

Note that if $x(t)$ has **odd** symmetry, then $x(0) = -x(0)$ so $x(0) = 0$.

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No! The property $x(0) = 0$ is a **necessary** condition for $x(t)$ to have odd symmetry, but it is not a **sufficient** condition.

Even and odd components

We can decompose any signal into even and odd components:

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) \triangleq \frac{1}{2} [x(t) + x(-t)], \quad x_o(t) \triangleq \frac{1}{2} [x(t) - x(-t)]$$

Question

Is the following $x(t)$ even or odd?

$$x(t) = \begin{cases} 1, & -1 < t < 3 \\ 0, & \text{otherwise,} \end{cases}$$

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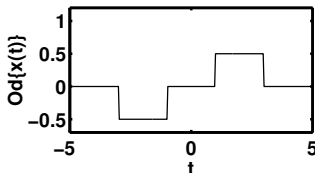
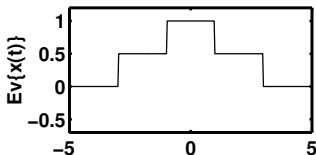
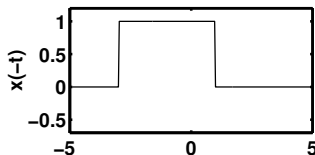
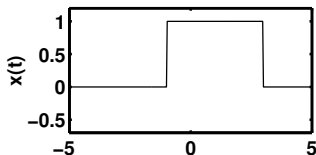
$$x(t) = \begin{cases} 1, & -1 < t < 3 \\ 0, & \text{otherwise,} \end{cases}$$

Neither. Find even and odd components.

Even and odd signals decomposition

$$x_e(t) = \begin{cases} 1/2, & -3 < t \leq -1 \\ 1, & -1 < t < 1 \\ 1/2, & 1 \leq t < 3 \\ 0, & \text{otherwise,} \end{cases}$$

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Average value and energy

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The **average value** of a signal is defined as

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Example

The average value of an odd signal is zero.

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If E is finite ($E < \infty$) then $x(t)$ is called an **energy signal** and $P = 0$.

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Energy and power signals

Definition

If E is infinite, then P can be either finite or infinite. If P is finite and nonzero, then $x(t)$ is called a **power signal**.

Some signals are neither energy signals nor power signals, such as $x(t) = t^2$, for which $E = \infty$ and $P = \infty$. Such signals are generally of little practical engineering importance.

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Energy and power signals: example

Example

consider $x(t) = 5 + a \cos t$ where $0 < a < \infty$.

- Find the average value of $x(t)$?
- Is $x(t)$ a power signal, an energy signal, or neither?

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Solution

The average value of this signal is:

$$\begin{aligned} A &\triangleq \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = \lim_{T \rightarrow \infty} \frac{1}{2T} (5t + a \sin t) \Big|_{-T}^T \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} (10T + 2a \sin T) = \lim_{T \rightarrow \infty} \left(5 - \frac{a \sin T}{2T} \right) = 5. \end{aligned}$$

Energy and power signals: solution

Solution

The average power of this signal is (by a similar integral):

$$\begin{aligned} P &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T (5 + a \cos t)^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T 25 + 10a \cos t + a^2 \left(\frac{1}{2} + \frac{1}{2} \cos 2t \right) dt \\ &= \lim_{T \rightarrow \infty} \frac{25(2T) + 10a(2 \sin T) + \frac{a^2}{2}(2T + \sin(2T))}{2T} \\ &= 25 + a^2/2. \end{aligned}$$

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*Since $0 < P < \infty$, $x(t)$ is a **power** signal.*

*Since P is nonzero and E is infinite, $x(t)$ is **not an energy** signal.*

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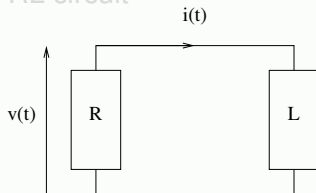
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Exponential signals (1)

Sinusoidal signals, exponential signals, and complex exponentials signals are particularly important because they arise from the solutions of linear constant-coefficient differential equations.

Example

an RL circuit



$$\frac{d}{dt}v(t) = \left(-\frac{R}{L}\right)v(t) = av(t)$$

where $a = -R/L$. Solution for $t > 0$ is

$$v(t) = v(0)e^{at} = v(0)e^{-t/\tau}$$

$$v(t) = L\frac{d}{dt}i(t) = L\frac{d}{dt}\left(-\frac{v(t)}{R}\right)$$

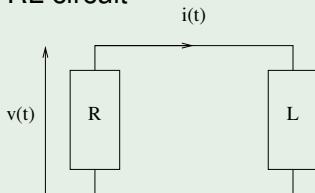
where $\tau = -1/a = L/R$ is called the **time constant** of the circuit. **(Picture)**

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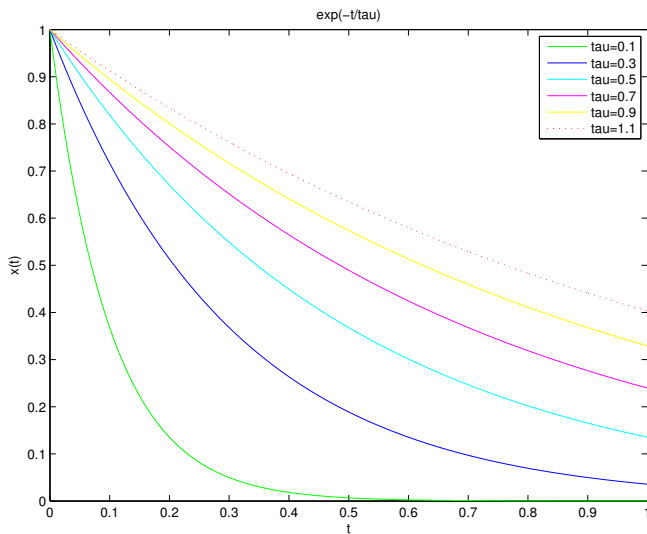
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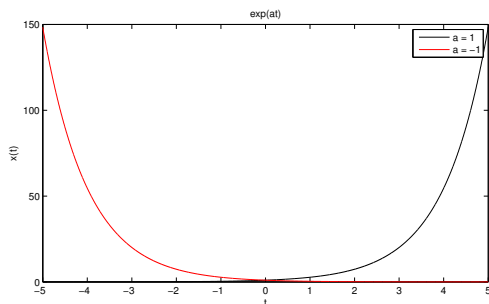
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Exponential signals (3)

Signals of the form $x(t) = ce^{at}$ are very important, for both real and complex c and a .

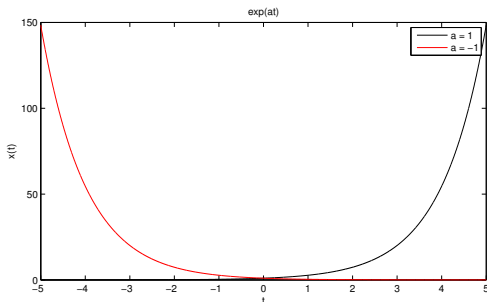
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Exponential signals (4)

- If a is purely imaginary, we get $x(t) = ce^{j\omega_0 t}$, a **complex exponential** signal.
- If $c = Ae^{j\phi}$, then $x(t) = Ae^{j\phi} e^{j\omega_0 t} = Ae^{j(\omega_0 t + \phi)} = A \cos(\omega_0 t + \phi) + jA \sin(\omega_0 t + \phi)$, a **sinusoid** signal.
- If $x(t) = e^{st}$ where $s = a + j\omega_0$ and $a < 0$, then $x(t) = e^{at}(\cos \omega_0 t + j \sin \omega_0 t)$ which is called a **damped sinusoid** signal. (See textbook p21.)

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Exponential signals (4)

- If a is purely imaginary, we get $x(t) = ce^{j\omega_0 t}$, a **complex exponential** signal.
- If $c = Ae^{j\phi}$, then $x(t) = Ae^{j\phi} e^{j\omega_0 t} = Ae^{j(\omega_0 t + \phi)} = A \cos(\omega_0 t + \phi) + jA \sin(\omega_0 t + \phi)$, a **sinusoid** signal.
- If $x(t) = e^{st}$ where $s = a + j\omega_0$ and $a < 0$, then $x(t) = e^{at}(\cos \omega_0 t + j \sin \omega_0 t)$ which is called a **damped sinusoid** signal. (See textbook p21.)

Euler's formula:

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

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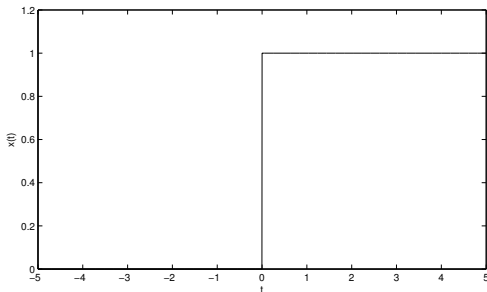
Unit step function/Signal

Definition

A **unit step function(signal)** is defined as

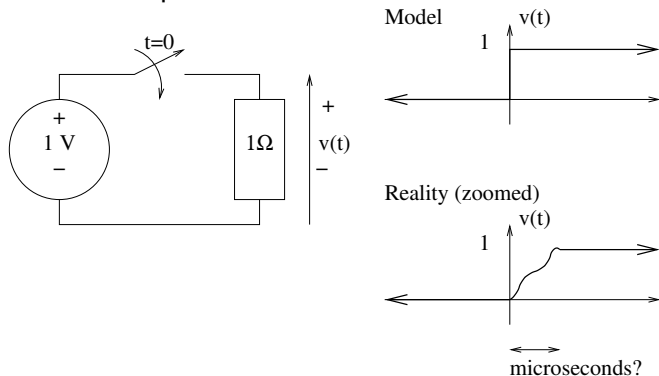
$$u(t) \triangleq \begin{cases} 1, & t > 0 \text{ or } t \geq 0 \\ 0, & t < 0 \end{cases}$$

The value at $t = 0$ is arbitrary and unimportant! Reasonable choices are 0, 1, and $\frac{1}{2}$; any will do.



Modeling a switch

The unit step is a useful model for a **switch**.

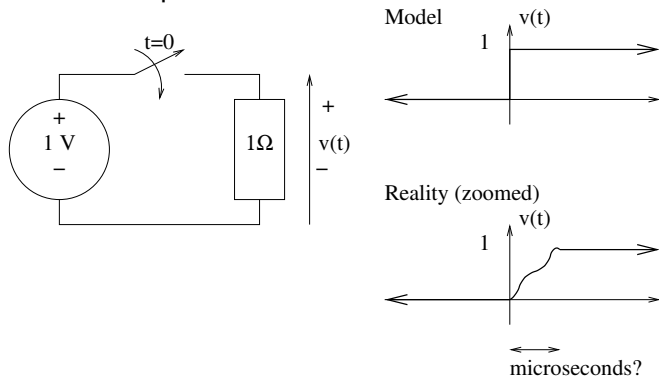


Question

- For a real switch in above is the voltage exactly a step function?
- Does the final voltage exactly equal 1 Volt?

Modeling a switch

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Question

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Modeling a switch (2)

Solution

- *No, as the first atoms of the switch contacts begin to “touch”, there will be a small current flow. As more atoms touch, more current will flow. Eventually the contacts will “touch completely”. and the system will settle to steady state.*
- *No, because of internal resistance of source and resistances of switch itself and wires.*

Modeling a switch (3)

- *But all of these effects are very small, so generally we can ignore them and the step function model $v(t) = u(t)$ is very reasonable for most purposes.*
- *In some high speed applications the switching time is important, so a more accurate model is necessary.*
- *Any real system can have no discontinuities of the type exhibited by the unit step function. There is always a small transition.*
- *It is OK to ignore this transition as long as the transition time is small relative to other time constants in the system being studied.*

Simplifying notation

The step function is useful for simplifying notation.

Example

The following step function “switches on” the signal from a guitar string plucked at time $t = 2$.

$$x(t) = \begin{cases} e^{-t} \sin(5t), & t > 2 \\ 0, & \text{otherwise} \end{cases} = e^{-t} \sin(5t) u(t - 2).$$

The first notation is messy, the second way is neat, and hides the braces within the definition of $u(t - 2)$.

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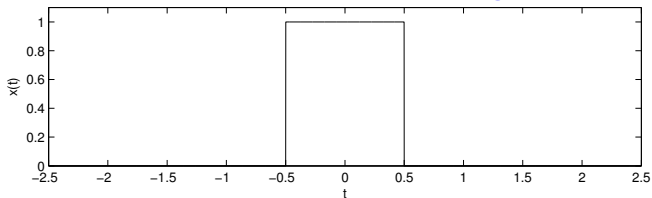
Rect(angle) function

Definition

A **rect(angle) function** is defined as

$$\text{rect}(t) \triangleq \begin{cases} 1, & -1/2 < t < 1/2 \\ 0, & \text{otherwise} \end{cases}$$

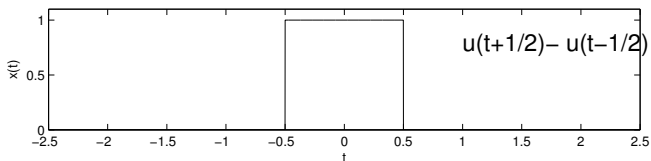
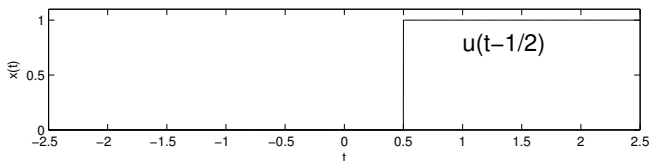
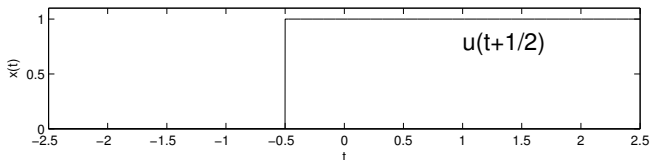
Centered at zero with **unit width and unit height**.



Rect function and step functions (1)

Can be represented using step functions:

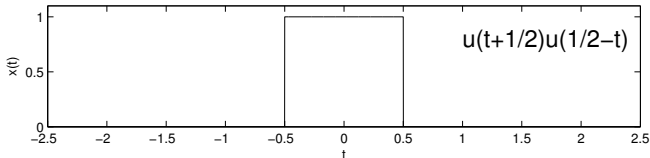
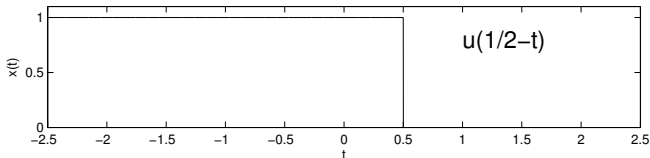
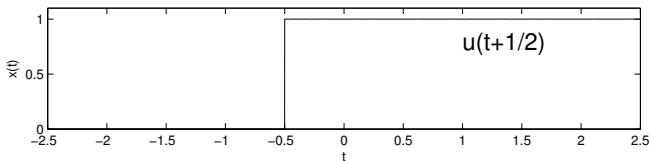
$$\text{rect}(t) = u(t + 1/2) - u(t - 1/2)$$



Rect function and step functions (2)

Can be represented using step functions:

$$\text{rect}(t) = u(t + 1/2)u(1/2 - t)$$

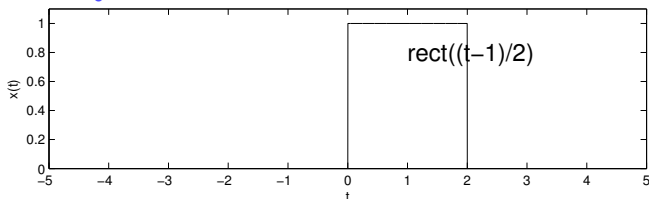


Transformed rect functions

Time-scaled and time-shifted rect function

$$\begin{aligned}\text{rect}\left(\frac{t-t_0}{T}\right) &= \begin{cases} 1, & -1/2 < \frac{t-t_0}{T} < 1/2 \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} 1, & t_0 - T/2 < t < t_0 + T/2 \\ 0, & \text{otherwise.} \end{cases}\end{aligned}$$

Centered at t_0 with width T

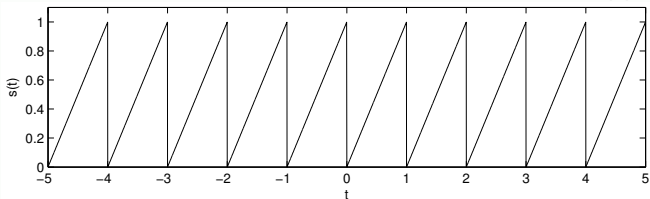


Rect function: example

Useful for “switching on and off” other functions, or for “extracting” on part of a signal, such as one period of a periodic signal.

Example

Find mathematical expression for a sawtooth signal $s(t)$.

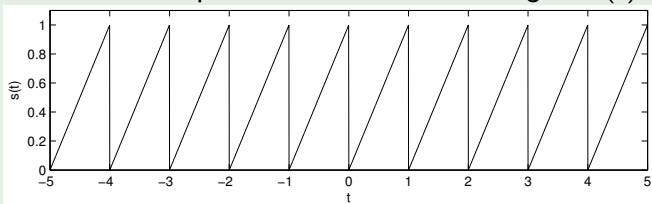


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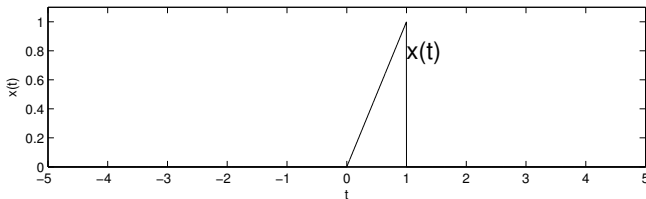
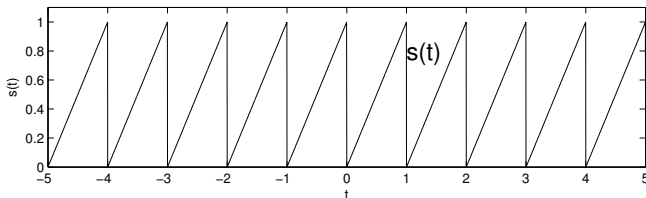
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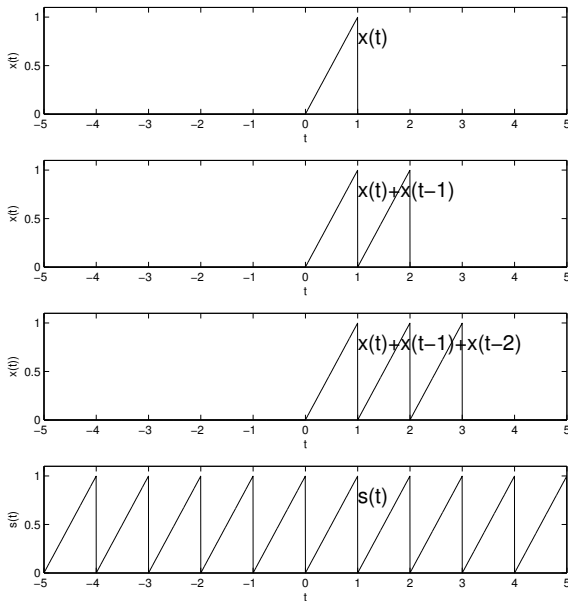
Rect function: solution (1)

Let $x(t)$ be the signal that is nonzero only over one period of $s(t)$ (one “tooth”).

$$x(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & \text{otherwise} \end{cases} = t \operatorname{rect}(t - 1/2).$$



Rect function: solution (2)



Rect function: solution (3)

$$s(t) = x(t) + x(t-1) + x(t-2) + \cdots$$

$$s(t) = \sum_{k=-\infty}^{\infty} x(t-k) = \sum_{k=-\infty}^{\infty} (t-k) \operatorname{rect}(t-k-1/2), \quad k \in \mathbb{Z}.$$

Very convenient closed-form representation. This form will be particularly useful for analyzing periodic signals using the Fourier series.

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Unit impulse function $\delta(t)$

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- It is another **mathematical idealization** that cannot occur in nature (like the unit step function), but is nevertheless useful for modeling certain phenomena, just as the step function is a useful idealization of a switch.
- More importantly, **it will greatly simplify our analysis of LTI systems later.**

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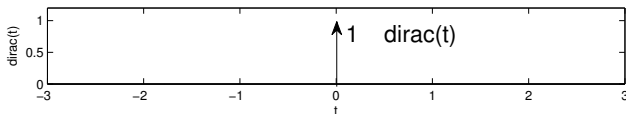
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Unit impulse function $\delta(t)$ (2)

$\delta(t)$ is like a pulse of zero width but infinite height and unit area.



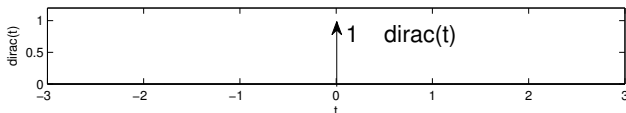
Graphical representation using upward arrow, labeled with area (called **weight**). (see text p.34 for scaled impulse.)

Such a thing is clearly not quite a “function” in the usual sense defined in calculus.

We can “define” an impulse function through its properties.

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We can “define” an impulse function through its **properties**.

Minor Properties

Property

- ① *unit area property* $\int_{-\infty}^{\infty} \delta(t - t_0) dt = 1$ for any t_0
- ② *scaling property* $\delta(at + b) = \frac{1}{|a|} \delta(t + b/a)$ for $a \neq 0$.
- ③ *symmetry property* $\delta(t) = \delta(-t)$
- ④ *support property* $\delta(t - t_0) = 0$ for $t \neq t_0$
- ⑤ *relationships with unit step function*: $\delta(t) = \frac{d}{dt} u(t)$,
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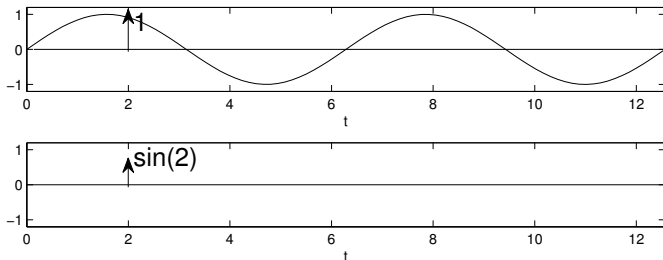
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Major properties: Sampling property

Property

Sampling property holds when $x(t)$ is continuous at t_0 :

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0).$$

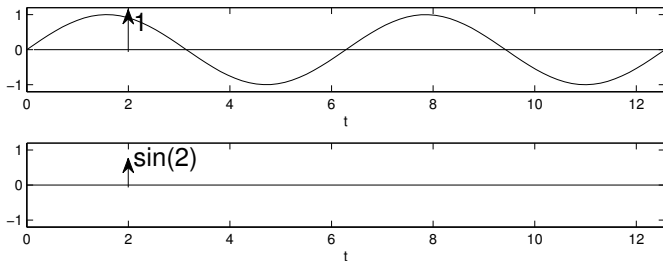


Major properties: Sifting property

Property

Sifting property holds when $x(t)$ is continuous at t_0 :

$$\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0).$$



Algebraic property

Property

Algebraic property

$$t\delta(t) = 0$$

Scaling property

Example

Show that $\delta(at) = \frac{1}{|a|}\delta(t)$ for $a \neq 0$.

Property

Let $g_1(t)$ and $g_2(t)$ be generalized functions. Then the **equivalence property** states that $g_1(t) = g_2(t)$ iff

$$\int_{-\infty}^{\infty} \phi(t)g_1(t) dt = \int_{-\infty}^{\infty} \phi(t)g_2(t) dt$$

for all suitably defined testing function $\phi(t)$.

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Scaling property: proof (1)

We prove this in two steps, for $a > 0$ and then for $a < 0$.

- When $a > 0$,*

$$\begin{aligned}\int_{-\infty}^{\infty} \phi(t) \delta(at) dt &= \int_{-\infty}^{\infty} \phi(t) \delta(|a| t) dt \\&= \int_{-\infty}^{\infty} \phi\left(\frac{1}{|a|} y\right) \delta(y) \frac{1}{|a|} dy \quad (y = |a| t \implies dy = |a| dt) \\&= \frac{1}{|a|} \phi\left(\frac{1}{|a|} y\right) \Big|_{y=0} = \boxed{\frac{1}{|a|} \phi(0)} \quad (\text{sifting property})\end{aligned}$$

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Scaling property: proof (2)

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Scaling property: proof (3)

Thus, for any a

$$\int_{-\infty}^{\infty} \phi(t) \delta(at) dt = \frac{1}{|a|} \phi(0)$$

Using the *sifting property*, we obtain

$$\frac{1}{|a|} \phi(0) = \frac{1}{|a|} \int_{-\infty}^{\infty} \phi(t) \delta(t) dt = \int_{-\infty}^{\infty} \phi(t) \frac{1}{|a|} \delta(t) dt$$

for any $\phi(t)$. Then, by the *equivalence property*, we obtain

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Ideal and practical impulse function

- *Mathematically, one can use the above properties to solve all the problems in this course.*
- *But physically we would like more insight, so we consider the “ideal” unit impulse function as a type of limit of “practical” impulse functions, just like the unit step function is a limit of practical almost-step functions.*

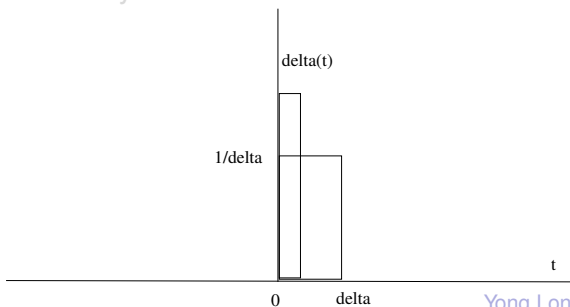
Practical impulse function

Definition

Practical impulse function, defined for any $\Delta > 0$:

$$\delta_{\Delta}(t) \triangleq \begin{cases} 1/\Delta, & 0 < t < \Delta \\ 0, & \text{otherwise.} \end{cases}$$

Note that area is **unity**, width approaches zero as $\Delta \rightarrow 0$; height approaches infinity as $\Delta \rightarrow 0$.



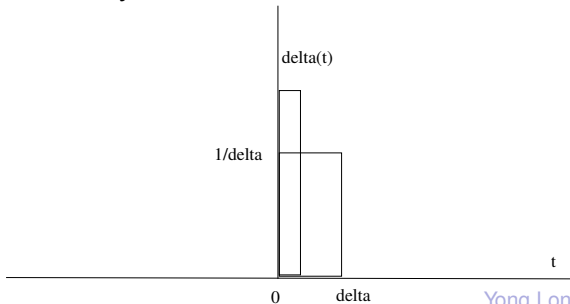
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Practical impulse function: example

Example

drumstick striking a drum (applied force vs time)

Example

metal hammer tapping a pendulum (applied force vs time)

Practical impulse function: example

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It is tempting to try to write

$$\delta(t) = \lim_{\Delta \rightarrow 0} \delta_{\Delta}(t)$$

but the limit is **not well defined mathematically**. Nevertheless, this is the intuition.

Instead we “define” $\delta(t)$ in terms of its **properties**, making sure that the properties are consistent with the above “limit”. Such objects are called **generalized functions** in mathematics.

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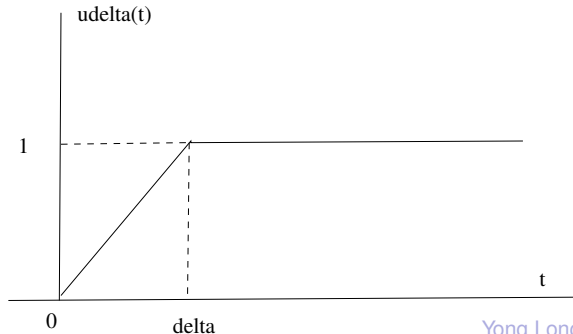
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Relationship to unit step function (1)

Explanation of $\delta(t) = \frac{d}{dt}u(t)$ using limiting step function.

Define a practical almost-step function as

$$u_{\Delta}(t) \triangleq \begin{cases} 0, & t \leq 0 \\ t/\Delta, & 0 < t < \Delta \\ 1, & t \geq \Delta \end{cases}$$

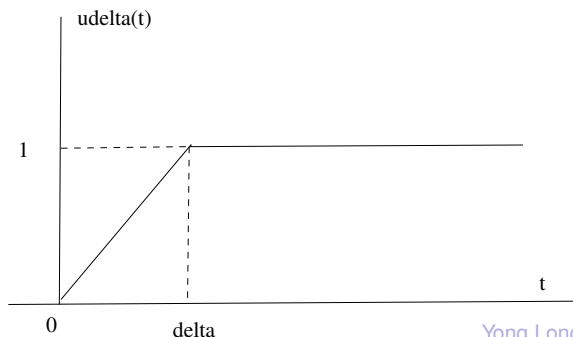


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Relationship to unit step function (2)

$$\frac{d}{dt}u_{\Delta}(t) = \begin{cases} 1/\Delta, & 0 < t < \Delta \\ 0, & \text{otherwise} \end{cases} = \delta_{\Delta}(t)$$

Then since $u(t) = \lim_{\Delta \rightarrow 0} u_{\Delta}(t)$, by taking the limit of both sides we have that

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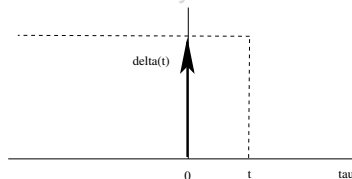
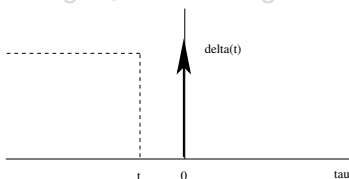
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Show graphically that $\int_{-\infty}^t \delta(\tau) d\tau = u(t)$.

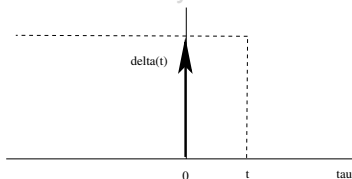
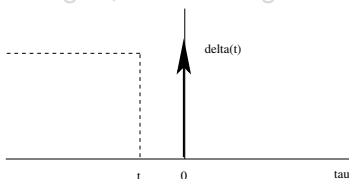
- For any $t < 0$, the range of integration over $(-\infty, t)$ will not cover zero, so the integral is simply zero.
- For any $t > 0$, integration over the range $(-\infty, t)$ covers zero, so the entire unit area of the impulse is included in the integral, so the integral evaluates to 1 for any $t > 0$.



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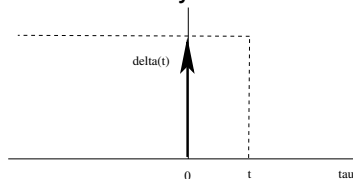
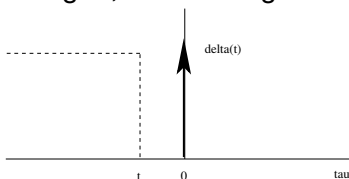
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Example (1)

Example

- By Newton's laws, velocity is the time-integral of acceleration.
- When a hammer taps a stationary pendulum, the pendulum (almost) instantaneously changes from being stationary to moving with some velocity that is related to how "hard" the hammer taps the pendulum.
- The acceleration is like a Dirac delta function, and the velocity is like a step function.

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Example

Let $x(t) = 2 \operatorname{rect}(t/2 - 3)$. Find $y(t) = \frac{d}{dt}x(t)$.

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Solution

$$x(t) = 2 \operatorname{rect}(t/2 - 3) = 2 \operatorname{rect}((t - 6)/2) = 2u(t - 5) - 2u(t - 7)$$

turns on at $t = 5$ and turns off at $t = 7$.

$$y(t) = \frac{d}{dt}x(t) = 2\delta(t - 5) - 2\delta(t - 7)$$

Example (3)

$$\begin{aligned}x(t) &= 2 \operatorname{rect}(t/2 - 3) = 2 \operatorname{rect}((t - 6)/2) \\&= 2u\left(\frac{t-6}{2} + \frac{1}{2}\right) - 2u\left(\frac{t-6}{2} - \frac{1}{2}\right) \\&= 2u\left(\frac{t-5}{2}\right) - 2u\left(\frac{t-7}{2}\right)\end{aligned}$$

directly plug in $\operatorname{rect}(t) = u(t + 1/2) - u(t - 1/2)$.

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Time-scaled unit step function

Question

$$u\left(\frac{t-t_0}{w}\right)? = u(t - t_0) \text{ where } w > 0$$

Time-scaled unit step function

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$u(\frac{t-t_0}{w})? = u(t-t_0)$ where $w > 0$ **Yes.**

- $u(\frac{t-t_0}{w})$ *turns on at t_0 .*

$$\frac{t-t_0}{w} \geq 0 \implies t \geq t_0$$

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Continuous-time systems

Definition

A **continuous-time system** is a device or process that, according to some well-defined rule, transforms one CT signal called the **input signal** or **excitation** into another CT signal called the **output signal** or **response**.

The input signal $x(t)$ is transformed by the system into a signal $y(t)$, which we express mathematically as

$$y(\cdot) = \mathcal{T}[x(\cdot)] \quad \text{or} \quad y(t) = \mathcal{T}[x(\cdot)](t) \quad \text{or} \quad x(\cdot) \xrightarrow{\mathcal{T}} y(\cdot).$$

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The notation $y(t) = \mathcal{T}[x(t)]$ is mathematically vague. Why?

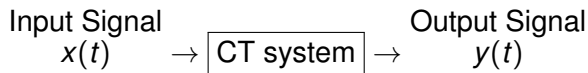
Notation

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The notation $y(t) = \mathcal{T}[x(t)]$ is mathematically vague. Why?

In general $y(t)$ is a function of the entire signal $x(\cdot)$ not just the single time point $x(t)$ as illustrated by auto example below.

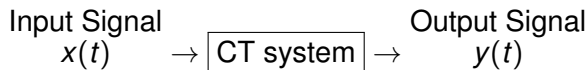
Diagram



The arrows in this diagram are not necessarily wires! They represent whatever medium transports the signal from one part of the system to another part.

At the systems level, we are less interested in the details of the implementation than in the **mathematical relationships** and **system properties**.

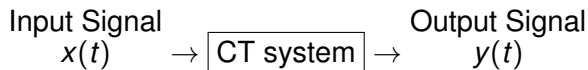
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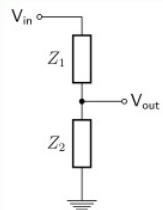
Example

voice (acoustic pressure) $\rightarrow$ microphone $\rightarrow$ electrical current

Example

voltage divider

(https://en.wikipedia.org/wiki/Voltage_divider)



For identical resistors, the output is $y(t) = \frac{1}{2}x(t)$. Called a **static** system.

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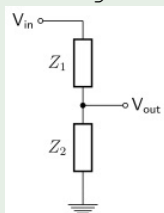
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Example (2)

Example

accelerator pedal position $\rightarrow$ engine/car $\rightarrow$ car velocity

- Input signal induces a response of the system.
- $x(t) = \frac{1}{2} + \frac{1}{2}u(t-3)$ (one pushes the gas pedal to the floor)
- $y(t) = 40 + 20(1 - e^{-t/\tau})$ (rise time or transient response)
- $y(t)$ is not solely a function of $x(t)$ at time t , but also a function of previous input signal values and the present and past state of the system.
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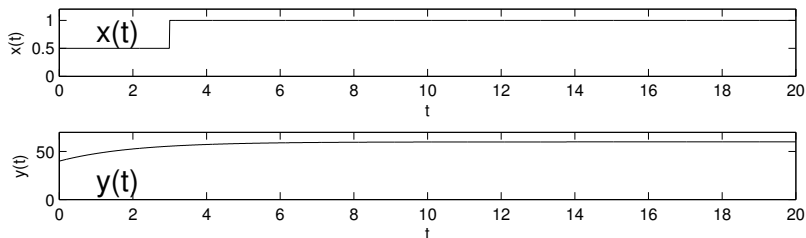
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$$x(t) = \frac{1}{2} + \frac{1}{2}u(t-3)$$

$$y(t) = 40 + 20(1 - e^{-t/\tau}), \quad \tau = 2$$



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Input-output description of systems (1)

Pictures/diagrams are a starting point, but for **quantitative analysis** every system must have an **input-output relationship**.

Definition

Input-output relationship is a mathematical expression that precisely defines how the output signal is related to the input signal.

Example

integrator system $y(t) = \int_{-\infty}^t x(\tau) d\tau$

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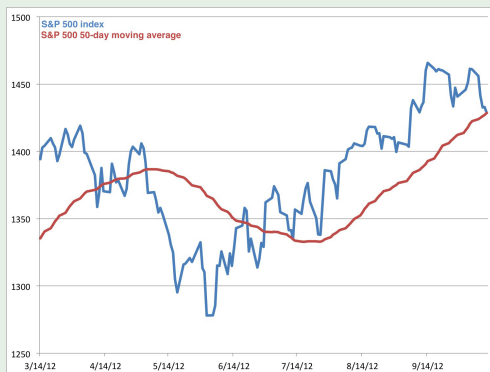
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Moving average

Example

moving average filter

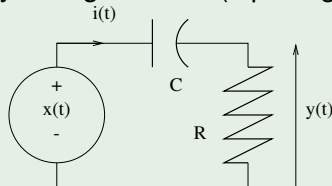
$$y(t) = \frac{1}{T} \int_{t-T}^t x(\tau) d\tau.$$



Input-output description of systems (2)

Example

RC circuit driven by voltage source (input signal).



$$\frac{1}{CR}y(t) + \frac{d}{dt}y(t) = \frac{d}{dt}x(t).$$

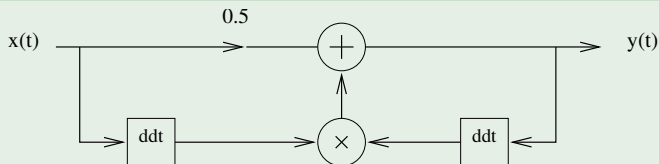
- This input-output relation is not of the form $y(t) = \text{some_function}[x(t)]$.
- When combined with an appropriate initial condition (such as 0 charge on the capacitor at time $t = 0$) one can solve the diffeq to determine $y(t)$ for any $x(t)$ (**later lectures.**)

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Block diagram representation of CT systems

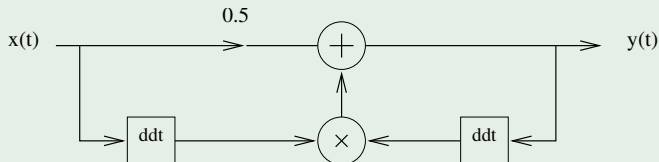
Example



The lower-right part is called a **feedback connection**.

Block diagram representation of CT systems

Example



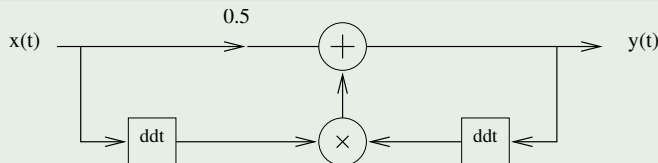
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Basic elements:

- **adder** (see text Fig. 2.29 on P. 126)
- **constant multiplier (amplifier)** (see text Fig. 2.29 on P. 126)
- **signal multiplier**
- **differentiator** (see text Fig. 2.29 on P. 126)
- **integrator** (see text Fig. 2.31 & 2.32 on P. 127)

Block diagram representation of CT systems

Example



The lower-right part is called a **feedback connection**.

Input-output relationship defined by the diagram:

$$y(t) = 0.5x(t) + \left(\frac{d}{dt}x(t) \right) \cdot \left(\frac{d}{dt}y(t) \right).$$

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Interconnection of systems

1 Series connection

$$x(t) \rightarrow \boxed{\mathcal{T}_1} \rightarrow \boxed{\mathcal{T}_2} \rightarrow y(t)$$

Mathematically: $y(t) = \mathcal{T}_2[\mathcal{T}_1[x(t)]]$.

2 Parallel connection

(See text Fig. 1.42(b) on P. 42)

Mathematically: $y(t) = \mathcal{T}_1[x(t)] + \mathcal{T}_2[x(t)]$.

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Example

Example

$x(t) \rightarrow$ amplifier, gain=5 $\rightarrow$ differentiator $\rightarrow y(t)$

$$y(t) = 5 \frac{d}{dt} x(t)$$

In this example the order of interconnection is irrelevant. We will learn soon that this is because both subsystems are linear.

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Classification of CT systems

Two general aspects to categorize:

- **Amplitude** properties
 - A-1 linearity (1.6.6)
 - A-2 stability (1.6.4)
 - A-3 invertibility (1.6.2)
- **Time** properties
 - T-1 causality (1.6.3)
 - T-2 memory (1.6.1)
 - T-3 time-invariance (1.6.5)

Skill: Determining classifications of a given CT system

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Skill: *Determining classifications of a given CT system*

A-1 Linearity (1)

Definition

A system $\mathcal{T}$ is **linear** iff

$$\mathcal{T}[a_1 x_1(t) + a_2 x_2(t)] = a_1 \mathcal{T}[x_1(t)] + a_2 \mathcal{T}[x_2(t)]$$

for **any signals** $x_1(t)$, $x_2(t)$ and **any (even complex) constants** a_1 and a_2 . Otherwise the system is called **nonlinear**.

Response to a weighted sum of input signals is the weighted sum of the individual responses. (*Picture*)

A-1 Linearity (1)

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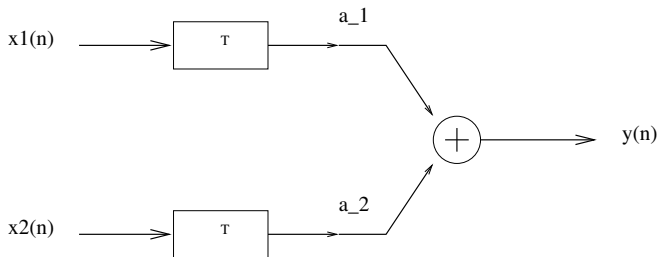
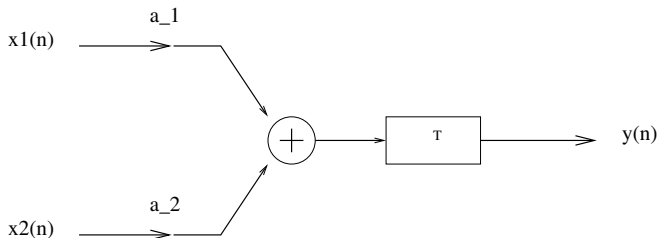
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A-1 Linearity (2)



A-1 Linearity (3)

Question

We will focus on linear systems. Why?

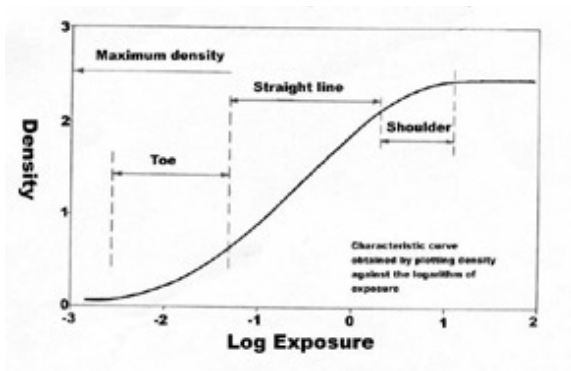
A-1 Linearity (3)

Question

We will focus on linear systems. Why?

- 1 The class of linear systems is *easier to analyze*.
- 2 Often *linearity is desirable* - avoids distortions
(Example: amplifiers, audio mixers (superposition!)).
- 3 *Many nonlinear systems are approximately linear*
(Example: the characteristic curve of a photographic film relating the optical density of the film to the logarithm of the incident exposure. **(Picture)**).

A-1 Linearity (4)



Real systems are **never perfectly linear**, but often they are **approximately linear over an appropriate operating range**.

Two important special cases of linearity property (1)

Property

scaling property or homogeneity property:

$$\mathcal{T}[ax(t)] = a\mathcal{T}[x(t)]$$

Note that from $a = 0$ we see that **zero input signal implies zero output signal for a linear system.**

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Note that from $a = 0$ we see that **zero input signal implies zero output signal for a linear system.**

Two important special cases of linearity property (2)

Property

additivity property:

$$\mathcal{T}[x_1(t) + x_2(t)] = \mathcal{T}[x_1(t)] + \mathcal{T}[x_2(t)]$$

Using proof-by-induction, one can easily extend this property to the general superposition property

Property

general superposition property

$$\mathcal{T}\left[\sum_{k=1}^K x_k(t)\right] = \sum_{k=1}^K \mathcal{T}[x_k(t)].$$

In words: the response of a linear system to the sum of several signals is the sum of the response to each of the signals.

Two important special cases of linearity property (2)

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additivity property:

$$\mathcal{T}[x_1(t) + x_2(t)] = \mathcal{T}[x_1(t)] + \mathcal{T}[x_2(t)]$$

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- In general superposition need not hold for **infinite sums**; additional continuity assumptions are required.
- We assume the superposition summation holds even for infinite sums without further comment in this course.
- In fact, we even assume that superposition holds for **integrals**:

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Determining a system is linear or nonlinear

Skill: *Determining a system is linear or nonlinear.*

- 1 Find output signal $y_1(t)$ for a general input signal $x_1(t)$.
- 2 “Repeat” for input $x_2(t)$ and $y_2(t)$.
- 3 Find output signal $y(t)$ when input signal is $x(t) = a_1 x_1(t) + a_2 x_2(t)$.
- 4 If $y(t) = a_1 y_1(t) + a_2 y_2(t) \forall t$, then the system is linear.
- 5 If it does not appear that $y(t) = a_1 y_1(t) + a_2 y_2(t) \forall t$, then find a specific counter-example.

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Determining a system is linear or nonlinear

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- ① Find output signal $y_1(t)$ for a **general** input signal $x_1(t)$.
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- ③ Find output signal $y(t)$ when input signal is $x(t) = a_1x_1(t) + a_2x_2(t)$.
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- ⑤ If it does not appear that $y(t) = a_1y_1(t) + a_2y_2(t) \forall t$, then find a **specific counter-example**.

Example (1)

Example

Prove that the integrator is a linear system, where

$$y(t) = \int_{-\infty}^t x(\tau) d\tau.$$

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Prove that the integrator is a linear system, where

$$y(t) = \int_{-\infty}^t x(\tau) d\tau.$$

① $y_1(t) = \int_{-\infty}^t x_1(\tau) d\tau$

② $y_2(t) = \int_{-\infty}^t x_2(\tau) d\tau.$

③ *If the input is $x(t) = a_1 x_1(t) + a_2 x_2(t)$, then the output is*

$$\begin{aligned} y(t) &= \int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^t [a_1 x_1(\tau) + a_2 x_2(\tau)] d\tau \\ &= a_1 \int_{-\infty}^t x_1(\tau) d\tau + a_2 \int_{-\infty}^t x_2(\tau) d\tau = a_1 y_1(t) + a_2 y_2(t). \end{aligned}$$

④ *Since this holds for all t , for all input signals $x_1(t)$ and $x_2(t)$, and for any constants a_1 and a_2 , the integrator is **linear**.*

Example (2)

Example

Determine whether linearity holds for $y(t) = \int_{-\infty}^t x^3(\tau) d\tau$.

Example (2)

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Determine whether linearity holds for $y(t) = \int_{-\infty}^t x^3(\tau) d\tau$.

Direct method:

① $y_1(t) = \int_{-\infty}^t x_1^3(\tau) d\tau$

② $y_2(t) = \int_{-\infty}^t x_2^3(\tau) d\tau$.

③ *If the input is $x(t) = a_1 x_1(t) + a_2 x_2(t)$, then the output is*

$$\begin{aligned} y(t) &= \int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^t [a_1 x_1(\tau) + a_2 x_2(\tau)]^3 d\tau \\ &\neq a_1 y_1(t) + a_2 y_2(t). \end{aligned}$$

Example (2)

Example

Determine whether linearity holds for $y(t) = \int_{-\infty}^t x^3(\tau) d\tau$.

Counter-example:

- if $x(t) = u(t)$, then $y(t) = tu(t)$, which is called the **unit ramp signal**.
- But if $x(t) = 2u(t)$, then $y(t) = 8tu(t)$, so doubling the input did not double the output.
- Thus the **scaling property is violated**, so this system is **nonlinear**.

Example (3)

Example

Are the following systems linear?

- $y(t) = x^3(t)$
- $y(t) = 2x(t) + 3$

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-
- *To show that a signal is **nonlinear**, all that is needed is **a counter-example to the scaling and additivity properties**.*
 - *The scaling property will usually (but not always!) suffice.*

Example (3)

Example

Are the following systems linear?

- $y(t) = x^3(t)$
- $y(t) = 2x(t) + 3$

- *Let $x_1(t) = 1$, a constant signal. Then $y_1(t) = 1$.*
- *Now suppose the input is $x(t) = 2x_1(t) = 2$, then the output is $y(t) = 2^3 = 8 \neq 2y_1(t) = 2$*
- *So the system is **nonlinear**.*

Example (3)

Example

Are the following systems linear?

- $y(t) = x^3(t)$
- $y(t) = 2x(t) + 3$

nonlinear, a zero input yields the output $y(t) = 3$, which is nonzero.

Example (4)

Example

Is $y(t) = \text{Real}[x(t)]$ linear?

Example (4)

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Is $y(t) = \text{Real}[x(t)]$ linear?

*For a system to be called linear, we require the **superposition property** to hold even for **complex signals and scaling constants**.*

Example (4)

Example

Is $y(t) = \text{Real}[x(t)]$ linear?

*$y(t) = \text{Real}[x(t)]$ satisfies additive property, but not scaling property for complex a . Thus it is **not linear**.*

A-2 Stability (1)

Definition

A system is **bounded-input bounded-output (BIBO) stable** iff every bounded input produces a bounded output.

If $\exists M_x$ s.t. $|x(t)| \leq M_x < \infty \forall t$, then there must exist an M_y s.t.
 $|y(t)| \leq M_y < \infty \forall t$.

Usually M_y will depend on M_x .

Otherwise the system is called **unstable**, and it is possible that a small input signal will make the output “blow up.”

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Is the integrator system $y(t) = \int_{-\infty}^t x(\tau) d\tau$ BIBO stable?

A-2 Stability: example (1)

Example

Is the integrator system $y(t) = \int_{-\infty}^t x(\tau) d\tau$ BIBO stable?

*Consider input signal $x(t) = u(t)$, which is bounded by $M_x = 1$. But $y(t) = tu(t)$ blows up, so the integrator is an **unstable** system.*

Triangle inequality

The **triangle inequality** is sometimes useful for proving that a system is BIBO stable.

- $|a + b| \leq |a| + |b|$ (Easily proved by considering 4 cases where a and b are positive or negative.)
- $|\sum_n a_n| \leq \sum_n |a_n|$
- $|\int f(t) dt| \leq \int |f(t)| dt$

A-2 Stability: example (2)

Example

Is the moving average $y(t) = \frac{1}{T} \int_{t-T}^t x(\tau) d\tau$ for $T > 0$ system BIBO stable?

A-2 Stability: example (2)

Example

Is the moving average $y(t) = \frac{1}{T} \int_{t-T}^t x(\tau) d\tau$ for $T > 0$ system BIBO stable?

Suppose $|x(t)| \leq M_x < \infty \forall t$, so $x(t)$ is a bounded input. Then by the triangle inequality:

$$|y(t)| \leq \frac{1}{T} \int_{t-T}^t |x(\tau)| d\tau \leq \frac{1}{T} \int_{t-T}^t M_x d\tau = M_x,$$

*so the output signal is also bounded for a bounded input. Thus the moving average system is **BIBO stable**.*

A-2 Stability: example (3)

Example

Is $y(t) = x^5(t)$ BIBO stable?

A-2 Stability: example (3)

Example

Is $y(t) = x^5(t)$ BIBO stable?

*Suppose $|x(t)| \leq M_x < \infty$. Then $|y(t)| = |x^5(t)| \leq M_x^5 < \infty$. So this system is **BIBO stable**.*

A-3 Invertibility

Definition

A system $\mathcal{T}$ is called **invertible** iff each (possible) output signal is the response to only one input signal. Otherwise $\mathcal{T}$ is not **invertible**.

Property

If a system $\mathcal{T}$ is invertible, then there exists a system $\mathcal{T}^{-1}$ such that

$$x(t) \rightarrow \boxed{\mathcal{T}} \rightarrow y(t) \rightarrow \boxed{\mathcal{T}^{-1}} \rightarrow z(t) = x(t).$$

Mathematically:

$$\mathcal{T}^{-1}[\mathcal{T}[x(t)]] = x(t)$$

Design of $\mathcal{T}^{-1}$ is important in many signal processing applications.

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A-3 Invertibility: example (1)

Example

encryption/decryption for secure communication. Needs to be invertible for no loss of information.

Example

digital speedometer

velocity $\rightarrow$ speed sensor $\rightarrow$ voltage

$\rightarrow$ mathematical inverse of sensor law $\rightarrow$ velocity display.

We display the velocity, not the voltage, so there should be a one-to-one relationship between the two.

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A-3 Invertibility: example (2)

Example

Is the full-wave rectifier: $y(t) = |x(t)|$ invertible?

A-3 Invertibility: example (2)

Example

Is the full-wave rectifier: $y(t) = |x(t)|$ invertible?

*The distinct input signals $x(t) = \sin t$ and $x(t) = -\sin t$ yield the same output signal. So **not invertible**.*

A-3 Invertibility: example (3)

Example

Is the exponential-law device: $y(t) = e^{x(t)}$ invertible?

A-3 Invertibility: example (3)

Example

Is the exponential-law device: $y(t) = e^{x(t)}$ invertible?

Invertible (for real input signals anyway). Inverse system is $x(t) = \log y(t)$, a log-law device.

A-3 Invertibility: example (4)

Example

Is the ideal amplifier: $y(t) = 2x(t)$ invertible?

A-3 Invertibility: example (4)

Example

Is the ideal amplifier: $y(t) = 2x(t)$ invertible?

Invertible. Inverse system is $x(t) = \frac{1}{2}y(t)$.

T-1 Causal systems

Definition

For a **causal** system, the output $y(t)$ at any time t depends **only** on the “present” and (possibly) “past” inputs *i.e.* on $x(t)$ and on various $x(t_0)$ for $t_0 \leq t$ only, but **not** on future inputs. Otherwise **noncausal** system.

Causality is necessary for **real-time implementation**. Noncausal systems arise primarily when t is some other variable than time, such as space.

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T-1 Causal systems: example (1)

Example

Is the integrator: $y(t) = \int_{-\infty}^t x(\tau) d\tau$ causal?

T-1 Causal systems: example (1)

Example

Is the integrator: $y(t) = \int_{-\infty}^t x(\tau) d\tau$ causal?

Solution

*It only depends on values of x for $\tau \leq t$, so **causal**.*

T-1 Causal systems: example (2)

Example

Is the symmetric moving average: $y(t) = \frac{1}{2T} \int_{t-T}^{t+T} x(\tau) d\tau$ causal? (Useful for image processing.)

T-1 Causal systems: example (2)

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Solution

noncausal. It depends on $t + T \geq \tau > t$

T-2 Memory

Definition

For a **static system** or **memoryless** system, the output $y(t)$ depends only on the current input $x(t)$, not on previous or future values of the input signal.

Otherwise it is a **dynamic system** and must have memory.

Example

- $y(t) = e^{x(t)} / \sqrt{|t+3|}$.
- moving average $y(t) = \frac{1}{T} \int_{t-T}^t x(\tau) d\tau, T > 0$

Dynamic systems are the interesting ones and will be **our focus**. (This time we take the more complicated choice!)

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Dynamic systems are the interesting ones and will be **our focus**. (This time we take the more complicated choice!)

Memory vs. causality

Question

- *Is a memoryless system necessarily causal?*
- *Is a dynamic system necessarily noncausal?*

Memory vs. causality

Question

- *Is a memoryless system necessarily causal?* **Yes**
- *Is a dynamic system necessarily noncausal?* **No**. Dynamic systems can be causal or noncausal.

Memory vs. causality

Question

- *Is a memoryless system necessarily causal?*
- *Is a dynamic system necessarily noncausal?*

Memory is often associated with stored energy, such as the charge on a capacitor, or the kinetic energy of a moving object.

T-3 Time-invariance (1)

Systems whose input-output behavior does not change with time are called **time-invariant** will be **our focus**.

- “**Easier**” to analyze.
- Time-invariance is a **desired property** of many systems.

We will focus primarily, but not exclusively, on time-invariant systems.

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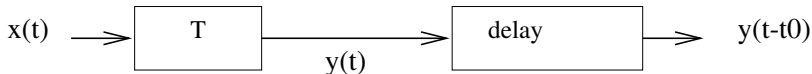
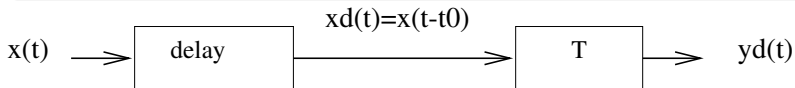
T-3 Time-invariance (2)

Definition

A system $\mathcal{T}$ is called **time invariant** or **shift invariant** iff

$$x(t) \xrightarrow{\mathcal{T}} y(t) \text{ implies that } x(t - t_0) \xrightarrow{\mathcal{T}} y(t - t_0)$$

for **every** input signal $x(t)$ and time shift t_0 . Otherwise the system is called **time variant** or **shift variant**.



Recipe for showing time-invariance

Recipe for showing time-invariance

- 1 Determine output signal $y(t)$ due to a generic input signal $x(t)$.
- 2 Determine the **delayed output** signal $y(t - t_0)$, by **replacing t with $t - t_0$ in $y(t)$ expression**.
- 3 Determine output signal $y_d(t)$ due to a **delayed input** signal $x_d(t) = x(t - t_0)$.
- 4 If $y_d(t) = y(t - t_0)$, then system is time-invariant.

Time-invariance: example (1)

Example

Is the symmetric moving average filter

$y(t) = \frac{1}{3}[x(t-1) + x(t) + x(t+1)]$ time-invariant?

Time-invariance: example (1)

Example

Is the symmetric moving average filter

$$y(t) = \frac{1}{3}[x(t-1) + x(t) + x(t+1)] \text{ time-invariant? } \text{Yes}$$

① Output due to $x(t)$ is $y(t) = \frac{1}{3}[x(t-1) + x(t) + x(t+1)]$

② Delayed output is

$$y(t-t_0) = \frac{1}{3}[x(t-t_0-1) + x(t-t_0) + x(t-t_0+1)]$$

③ Output due to *delayed input* $x_d(t) = x(t-t_0)$ is

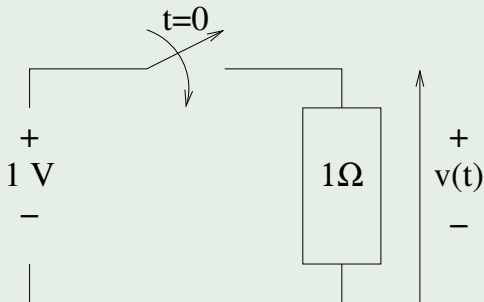
$$\begin{aligned} y_d(t) &= \frac{1}{3}[x_d(t-1) + x_d(t) + x_d(t+1)] \\ &= \frac{1}{3}[x(t-t_0-1) + x(t-t_0) + x(t-t_0+1)] \end{aligned}$$

④ $y_d(t) = y(t-t_0)$

Time-invariance: example (2)

Example

A switch that closes at $t = 1$.



- 1 How to represent input-output relationship mathematically?
- 2 Is it Time invariant? If no, find a counter-example.

Time-invariance: solution (2)

Solution

- 1 $y(t) = u(t - 1)x(t)$.
- 2 No. *Time-varying gain* $u(t - 1)$.
- 3 A counter-example.
 - If $x(t) = \delta(t)$, then $y(t) = 0$.
 - But if $x_d(t) = \delta(t - 2)$, then
$$y_d(t) = \delta(t - 2)u(t - 1) = \delta(t - 2) \neq y(t - 2).$$

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Time-invariance: solution (3)

Example

A switch that closes at $t = 1$: $y(t) = u(t - 1)x(t)$.

For certain t_0 , $y_d(t) = y(t - t_0)$.

if $x_d(t) = \delta(t + 2)$, then $y_d(t) = \delta(t + 2)u(t - 1) = 0 = y(t + 2)$.

A time-varying gain results in a time-varying system while systems with constant gains are time-invariant (e.g., $y(t) = 2x(t)$).

Time-invariance: solution (3)

Example

A switch that closes at $t = 1$: $y(t) = u(t - 1)x(t)$.

For certain t_0 , $y_d(t) = y(t - t_0)$.

if $x_d(t) = \delta(t + 2)$, then $y_d(t) = \delta(t + 2)u(t - 1) = 0 = y(t + 2)$.

A time-varying gain results in a time-varying system while systems with constant gains are time-invariant (e.g., $y(t) = 2x(t)$).

Time-invariance: example (3)

Example

Is the modulator $y(t) = \cos(\pi t) x(t)$ time-invariant?

Time-invariance: example (3)

Example

Is the modulator $y(t) = \cos(\pi t) x(t)$ time-invariant?

No.

Time-invariance: example (3)

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No. How do we show lack of a property? Find a counter-example.

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$$\text{If } x(t) = \delta(t) \quad \text{then} \quad y(t) = \delta(t)$$

$$\text{If } x_d(t) = \delta(t-1) \quad \text{then} \quad y_d(t) = -\delta(t-1) \neq \delta(t-1) = y(t-1)$$

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The modulator is a useful system! (AM radio)

Time-invariance: example (4)

Example

Is the amplified time reversal $y(t) = 3x(-t)$ time-invariant?

Time-invariance: example (4)

Example

Is the amplified time reversal $y(t) = 3x(-t)$ time-invariant?

Solution

No. Any time shift in the input will be time-reversed.

$$\begin{aligned} x(t) = \delta(t) \xrightarrow{\mathcal{T}} y(t) &= 3x(-t) = 3\delta(-t) \\ &= 3\delta(t) \text{ (by symmetry property of unit impulse)} \end{aligned}$$

$$\begin{aligned} x_d(t) = \delta(t-1) \xrightarrow{\mathcal{T}} y_d(t) &= 3x_d(-t) = 3x_d(z) \quad (\text{where } z = -t) \\ &= 3\delta(z-1) = 3\delta(-t-1) = 3\delta(-(t+1)) \\ &= 3\delta(t+1) \neq 3\delta(t-1) = y(t-1), \end{aligned}$$

$y_d(t) \neq y(t-1)$ for $x_d(t) = \delta(t-1)$, so system is time varying.

Time-invariance: example (4)

Example

Is the amplified time reversal $y(t) = 3x(-t)$ time-invariant?

Any time shift in the input will be time-reversed. Similarly, time-scaled (compressed and expanded) systems are time-variant. (text Example 1.16)

Time-invariance: example (5)

Example

Is the integrator $y(t) = \int_{-\infty}^t x(\tau) d\tau$ time-invariant?

Time-invariance: example (5)

Example

Is the integrator $y(t) = \int_{-\infty}^t x(\tau) d\tau$ time-invariant?

Yes.

$$\textcircled{1} \quad y(t - t_0) = \int_{-\infty}^{t-t_0} x(\tau) d\tau$$

$$\textcircled{2} \quad x_d(t) = x(t - t_0)$$

$$\begin{aligned} y_d(t) &= \int_{-\infty}^t x_d(\tau) d\tau = \int_{-\infty}^t x(\tau - t_0) d\tau \\ &= \int_{-\infty}^{t-t_0} x(\tau') d\tau' \quad (\tau' = \tau - t_0) \\ &= y(t - t_0) \end{aligned}$$

Outline

- 1 1. Signals & Systems (Fundamentals)
 - Overview
 - Signal and System Definition
 - Classification of Signals
 - Signal Notation
 - Transformations of CT signals
 - Signal Characteristics
 - Exponential signals
 - Singularity functions (1.4)
 - Continuous-time systems
 - **Summary**

Summary (1)

- signal notation
- signal transformations
 - time transformations
 - amplitude transformations
 - differentiator / integrator systems
 - two-signal operations
- signal classes
 - even/odd signals
 - energy/power signals
 - periodic/apperiodic signals
 - exponential signals

Summary (2)

- singularity functions
 - unit step / rect signals
 - unit impulse function
 - impulse function properties (sifting, sampling, scaling)
- CT systems
- block diagrams
- system classes
 - amplitude properties: linearity, stability, invertibility
 - time properties: causality, memory, time-invariance

Key concepts/skills to study

Key concepts/skills to study

- time transformations
- braces/plots to rects/steps
- running integral operation
- properties of $\delta(t)$
- identifying signal properties
- identifying (all six) system properties